

Welding_M2 Exam Notes (Comprehensive, Self-Contained)

Source analyzed: Welding_M2.pdf (82 slides/pages)

These notes are built from all substantive content visible in the PDF. Repeated title/figure slides are consolidated to avoid duplication. Where a slide is diagram-heavy or ambiguous, it is clearly marked.

1) Course Context (Slides 1-4)

- Course: **Basic Manufacturing Processes (ME 204)**.
- Instructor: **Dr. Atul Singh Rajput** (NITK Surathkal).
- Broad modules listed:
 - Introduction to Manufacturing
 - Metal Casting
 - Metal Forming
 - Metal Joining
- Welding-focused coverage includes:
 - Introduction and classification of welding
 - Gas welding, arc welding, resistance welding, submerged arc welding
 - GMAW (MIG) and GTAW (TIG)
 - Carbon arc welding and advanced welding processes
 - Welding defects and inspection
 - Friction/friction stir welding, EBW, LBW, dissimilar metal joining
 - Welding codes, qualification, fatigue of welded joints
- Evaluation plan:
 - Internal: 20%
 - Mid-semester: 30%
 - End-semester: 50%

2) Manufacturing Basics and Classification (Slides 5-10)

2.1 What is Manufacturing?

- Manufacturing (from Latin roots) is conversion of **raw materials** into **useful products**.

2.2 Classification by Mass Change

- **Mass reducing** $((dM < 0))$: material removed.
- **Mass adding** $((dM > 0))$: material added.
- **Mass conserving** $((dM \approx 0))$: shape changed mainly without significant mass change.

2.3 Typical Process Families Mentioned

- Mass reducing:
 - Conventional machining: turning, milling, drilling, grinding
 - Non-conventional machining
- Mass adding:
 - Additive manufacturing
 - Powder metallurgy
- Mass conserving:
 - Casting, metal forming, joining processes

2.4 Why Non-conventional Machining?

- Machining **hard/ceramic materials**
- Producing **complex geometries** and **micro-features**

3) Introduction to Additive Manufacturing (Slides 11-21)

- Nature-inspired idea: near-net/low-waste manufacturing.
- Additive route presented conceptually as:
 - **CAD model -> slicing -> final product.**
- Emphasis: additive processes can reduce wastage compared to purely subtractive routes.

Note: Several slides in this block are mostly visual/illustrative with limited text.

4) Metal Joining Overview (Slides 22-23)

4.1 Why special joining for metals?

- Adhesives and common fasteners may not be adequate for many structural metal-joint demands.
- Heat-based metallurgical methods are highlighted: **soldering, brazing, welding.**

4.2 Joining Process Taxonomy

- **Metallurgical** joining: welding, brazing, soldering
- **Mechanical** joining: screws/fasteners/riveting/compression/folding etc.
- **Adhesive** joining: natural/elastomer/thermoplastic/thermoset systems
- Also classified as:
 - **Permanent (P)**
 - **Non-permanent (NP)**

5) Welding Fundamentals (Slides 24-26)

5.1 Definition and Principle

- Welding is a **permanent joining process** where metals are joined by heating (and sometimes pressure) to create coalescence.
- Filler may be:
 - supplied by electrode itself, or
 - supplied separately.
- Similar and dissimilar metals may be joined.
- Typical stated high-temperature range in arc context: about **6000 to 7000 degC**.

5.2 Heat Sources for Welding

1. Chemical reaction
2. Electric arc
3. Electrical resistance
4. Frictional heat

5.3 Broad Welding Classification

- **Plastic/pressure welding:**
 - Metals heated to plastic state and forced together by pressure
 - generally without filler
 - examples: forge welding, resistance welding
- **Fusion welding:**
 - no pressure required
 - joint region melted and resolidified
 - filler may or may not be added
 - examples: oxy-acetylene, carbon arc welding

6) Additional Classification Labels (Slides 27-36, repeated)

Repeated slides present category labels:

- **Autogenous**
- **Homogenous**
- **Heterogeneous**
- Associated with solid/liquid/mixed-state wording in the deck.

Ambiguity note: These slides are mostly diagrammatic/repetitive and contain minimal explanatory text.

7) Weld Types, Joints, and Arc Welding Basics (Slides 37-40)

7.1 Welded Joints

- Slides introduce **types of welds and joints** and examples (visual-heavy).

7.2 Arc Welding Principle

- Fusion welding method where arc heat melts the joint region.
- Arc exists between electrode (anode/cathode role shown) and base metal.
- Electrical energy -> thermal energy.
- Arc temperature cited around **6000 to 7000 degC**.

7.3 Electrodes

- **Non-consumable**: carbon/graphite/tungsten.
- **Consumable**: typically steel wire/rod; may be bare or flux-coated.
- Flux-coated electrode protection mechanisms (SMAW context):
 - slag cover over weld pool
 - shielding gases generated by flux decomposition

8) Arc Welding Processes (Slides 41-53)

8.1 Manual Metal Arc Welding (SMAW) (slide 42 + figure slides)

- Consumable flux-coated electrode provides filler.
- High localized heat input; narrow HAZ relative to heat concentration statement.
- Widely used due to versatility and low setup cost.
- Manual process -> quality strongly depends on welder skill.
- Electrode dimensions mentioned: approx. diameter 1.6 to 6.3 mm; length 250 to 450 mm.
- Power sources:
 - AC transformer
 - DC generator/rectifier
- Suitable for wide plate-thickness range and many commercial metals/alloys.
- Usable in all welding positions.

8.2 Gas Metal Arc Welding (GMAW/MIG) (slides 45-46)

- Continuous consumable wire acts as current carrier + filler.
- Arc between wire and workpiece.
- External shielding gas: Ar, CO₂, or mixtures.
- Common applications: automobile, sheet-metal fabrication, pressure vessels.
- Suitable thickness: roughly **1 mm to 25 mm**.
- Arc temperature: roughly **6500 to 7500 degC**.
- Typical source: **CV DC**, usually **DCEP**.

8.3 Gas Tungsten Arc Welding (GTAW/TIG) (slides 47-48)

- Arc between non-consumable tungsten electrode and workpiece.
- Filler added separately when needed.
- Shielding by inert gas (argon/helium).

- Precision applications: aerospace, nuclear, chemical sectors.
- Suitable especially for thin sections: approx. **0.5 to 6 mm** (up to ~10 mm by multipass).
- Arc temperature can reach ~**11,000 degC**.
- Typical source: **CC**;
- AC for Al/Mg
- DC for steels/other metals.

8.4 Submerged Arc Welding (SAW) (slides 49-50)

- Arc and weld pool submerged under granular flux blanket.
- Benefits: reduced spatter, reduced arc radiation, good pool protection.
- Used for thick plates: approx. **10 mm to 100 mm**.
- Arc temperature: ~**7000 to 8000 degC**.
- Source: **CV**, AC or DC depending on use.
- Typical applications: pressure vessels, large pipes, tanks, ship hulls.

8.5 Plasma Arc Welding (PAW) (slides 51-52)

- Constricted plasma arc through narrow nozzle -> high energy density.
- High precision and repeatability.
- Suitable thickness: roughly **0.1 mm to 8 mm**.
- Plasma arc temperature: about **16,000 to 20,000 degC**.
- Power source: **CC DC**.
- Applications: aerospace, medical devices, precision industries.

8.6 Arc Welding: Advantages, Disadvantages, Applications (slide 53)

Advantages

- Simple setup, portable equipment
- Can weld most metals/alloys
- Good weld quality achievable

Disadvantages

- Slower process (in context of production alternatives)
- Post-cleaning needed (slag removal)

Applications

- Tanks, boilers, pressure vessels, shipbuilding, pipe joining

9) Metal Transfer Modes and Power Source Logic (Slides 54-58)

9.1 Metal Transfer Modes

- Short circuit
- Globular

- Spray
- Pulse spray
- Rotating spray

9.2 Governing Forces Mentioned

- Electromagnetic pinch force
- Surface tension force
- Drag force
- Gravitational force

9.3 Deposition-Rate Table (as listed)

- Short circuit: **0.9 to 2.7 kg/hr**
- Globular: **1.8 to 3.2 kg/hr**
- Spray: **2.7 to 5.4 kg/hr**
- Pulse spray: **0.9 to 2.7 kg/hr**
- Rotating spray: **6.9 to 13.5 kg/hr**

9.4 Power Source Types

- **Constant Current (CC):** current stable, voltage varies with arc length
 - suited to manual arc-length processes (SMAW, GTAW)
- **Constant Voltage (CV):** voltage stable, current varies with wire-feed behavior
 - suited to wire-feed processes (MIG/MAG, SAW)

10) Wire Arc Additive Manufacturing (WAAM) (Slides 59-81)

10.1 Core WAAM Idea

- Wire feedstock melted by electric arc and deposited layer-by-layer.
- WAAM variants by arc process:
 - GMAW-based WAAM
 - GTAW-based WAAM
 - PAW-based WAAM
- High deposition rate stated: **~6 to 8 kg/hr** (vs many conventional AM methods).

10.2 Classification/Framework Slides

- Positioning by:
 - materials and material flow
 - machine kinematics (serial/parallel/hybrid references)
 - energy source (arc vs beam)
 - support mechanism
- Wire-beam AM (laser/e-beam) is contrasted with wire-arc AM.

10.3 Omnidirectionality and Vectors (slides 64-65)

- Mentions:
 - source-of-fusion vector
 - feedstock vector
 - deposition vector
- Directional relations and dynamic adjustment approaches are shown.
- Methods listed include coaxial or multiple feedstock / dynamic deposition direction control.

10.4 WAAM Setup and Materials (slide 66)

- Process setup/terminology slide for GMAW/GTAW/PAW systems.
- Materials listed include low-carbon steel, stainless steel, tool steel (P91), titanium alloys, aluminum alloys, Inconel, Stellite-type materials.

10.5 GMAW-Based WAAM Details (slides 67-76)

(a) Polarity/heat distribution concept

- Slides discuss DCEN, DCEP, and AC with different heat-distribution statements between electrode and substrate.
- Includes penetration/deposited bead visuals.

Ambiguity note: polarity labels and heat-split statements appear partly inconsistent on one slide; preserve with caution.

(b) Metal transfer modes reused in WAAM context

- Same mode names/rates as earlier arc-welding section.

(c) Cold Metal Transfer (CMT)

- Popular WAAM approach due to controlled heat input and reduced spatter.
- Attributed to Fronius (2003).
- Process cycle:
 1. Arc starts with main current and wire feeding.
 2. Droplet forms and wire dips into melt pool.
 3. Current is reduced for controlled transfer.
 4. Wire retracts after transfer.
 5. Main current reapplied; cycle repeats.

(d) Process parameters and experiment hierarchy (slides 71, 73)

- Parameter groups include:
 - Power/current/voltage (starting, main, end current)
 - Filler wire type/material/feed rate
 - Gas composition, flow rate, pressure, pre/post flow
 - Torch speed, stick-out, standoff

- Toolpath strategy and step-over
- Experimental levels:
 - **Single bead:** penetration, height, contact angle, profile
 - **Multi-bead:** step-over, flatness, layer-height control
 - **Multi-layer:** tensile, wear, hardness variation, fatigue/impact

(e) Bead-profile modeling (slide 72)

- Bead profile above substrate assumed symmetric parabola:

$$y = a + c x^2$$

with

$$a = h, \quad c = -\frac{4h}{w^2}$$

- Here, h = bead height, w = bead width.

(f) Low-carbon steel multilayer behavior

- Top layers may show higher hardness.
- Repeated thermal cycling can lower core hardness.
- Toolpath significantly affects hardness distribution.
- Heat treatment suggested for hardness uniformity.

(g) Tensile testing and trends

- ASTM E8 sub-size specimen reference.
- Samples should include:
 - deposition direction
 - build direction
 - transverse direction
- At least three samples suggested.
- Higher welding current -> higher tensile strength (via penetration), but trade-off with layer height/surface accuracy.

10.6 GTAW- and PAW-based WAAM (slides 77-79)

- GTAW WAAM: stainless steel/titanium deposition examples; DCEN/AC noted.
- PAW WAAM: equipment modules shown (chiller, wire feeders, plasma/welding unit, robotic setup).
- GTAW parameter matrix shown similar to GMAW parameter framework.

10.7 Gas effects and WAAM challenges (slides 80-81)

Gas effects (as listed)

- **Argon:** stable arc, smooth bead, oxidation control for non-ferrous materials.
- **CO₂:** low cost, better penetration, but more spatter.
- **Ar-CO₂ mix:** balanced stability/penetration with less spatter than pure CO₂.
- **Helium (often with Ar):** higher thermal conductivity -> higher heat input, penetration, and welding speed.

WAAM challenges

1. Surface-quality difficulty (heat distribution, arc instability, spatter buildup)
2. Residual stress/distortion due to thermal cycling
3. Material/feedstock limitations and weldability constraints
4. Need to minimize porosity, lack of fusion, and bead-shape irregularity
5. Support-structure design challenges for complex overhangs
6. Limited standardization/certification (emerging technology)

11) Slide Coverage Map (for completeness)

- Substantive text-heavy slides captured in notes:
 - 1-4, 6-10, 13-15, 21-26, 27/29/31/33/35, 39-40, 42, 45, 47, 49, 51, 53-55, 57, 60-81.
- Primarily title/figure/repeated slides (minimal new text):
 - 5, 11-12, 16-20, 28/30/32/34/36-38, 41, 43-44, 46, 48, 50, 52, 56, 58-59, 61-63, 65-66, 82.

12) Exam-Focused Preparation Section

12.1 High-yield definitions (must memorize)

1. Manufacturing
2. Welding (permanent metallurgical joining with/without pressure/filler)
3. Plastic welding vs fusion welding
4. Consumable vs non-consumable electrode
5. GMAW, GTAW, SAW, PAW
6. Constant current vs constant voltage source
7. WAAM and CMT

12.2 Important classifications/lists

- Manufacturing by mass change: reducing/adding/conserving.
- Welding heat sources: chemical/arc/resistance/friction.
- Metal transfer modes: short-circuit, globular, spray, pulse-spray, rotating spray.
- WAAM process classes: GMAW/GTAW/PAW.
- Gas effects: Ar vs CO₂ vs Ar-CO₂ vs He.

12.3 Key comparisons likely in exams

1. Plastic welding vs fusion welding
2. SMAW vs GMAW vs GTAW
3. SAW vs PAW
4. CC vs CV power source
5. GMAW-WAAM vs GTAW-WAAM (productivity vs precision style discussion)

12.4 Numerical/data points to retain

- Arc temperatures (approx):
 - Arc welding: 6000 to 7000 degC
 - GMAW: 6500 to 7500 degC
 - GTAW: up to ~11,000 degC
 - SAW: 7000 to 8000 degC
 - PAW: 16,000 to 20,000 degC
- Typical thickness ranges:
 - GMAW: 1 to 25 mm
 - GTAW: 0.5 to 6 mm (up to ~10 mm multipass)
 - SAW: 10 to 100 mm
 - PAW: 0.1 to 8 mm
- WAAM deposition rate: about 6 to 8 kg/hr (slide claim).
- Metal-transfer deposition rates: keep table values.

12.5 Formula to memorize

Bead profile model (single bead):

$$y = a + c x^2, \quad \text{quad } a=h, \quad \text{quad } c=-\frac{4h}{w^2}$$

where h = bead height and w = bead width.

12.6 Likely short-answer questions

1. Define welding and list its heat sources.
2. Distinguish consumable and non-consumable electrodes with examples.
3. Why is shielding gas needed in GMAW/TIG?
4. State advantages and disadvantages of arc welding.
5. List metal transfer modes in GMAW.
6. What is WAAM and why is it industrially attractive?
7. What is CMT and how does it reduce heat input/spatter?
8. Write differences between CC and CV power sources.

12.7 Likely long-answer questions

1. Explain and compare SMAW, GMAW, GTAW, SAW, and PAW (principle, power source, range, applications).
2. Discuss WAAM process flow, parameter groups, and quality challenges.
3. Explain bead geometry modeling and significance of width/height/penetration.
4. Discuss effect of polarity/current/gas/toolpath on WAAM outcomes.
5. Explain residual stress/hardness variation in multilayer WAAM and mitigation methods.

12.8 Safety/quality checkpoints to mention in answers

- Control heat input to avoid distortion/residual stress.
- Ensure proper shielding gas selection and flow settings.
- Prevent common defects: porosity, lack of fusion, irregular beads.
- Account for post-processing/cleaning (slag in relevant processes).
- Validate with mechanical tests (tensile/hardness/fatigue where required).

13) Ambiguities/Illegible Material Log

1. Several slides are image/diagram dominant, with limited extractable text; minor labels may not be fully readable.
2. One slide includes wording that appears inconsistent in polarity vs heat-distribution notation; reported as-is with caution.
3. One early content slide appears to contain a typo around friction/friction-stir wording.